RAG Learning

AQA A-Level Biology — Required Practicals

There are **12 required practicals** across the AQA A-Level Biology course. You will be tested on these in exams through questions about methods, apparatus, variables, analysis, evaluation, and the underlying biology. You do not need to have personally performed every practical, but you must understand them thoroughly.

Key practical skills tested: Independent, dependent and control variables; reliability (repeats, means); validity (controlling variables); accuracy and precision of measurements; sources of error and improvements; selecting appropriate apparatus; presenting data (tables, graphs); statistical analysis (chi-squared, t-test, Spearman rank correlation, standard deviation); drawing conclusions from data.

Table of Contents

1. RP1: Effect of a Variable on Enzyme-Controlled Reaction Rate
2. RP2: Stained Squashes of Root Tip Cells (Mitosis)
3. RP3: Dilution Series and Calibration Curve (Colorimetry)
4. RP4: Effect of a Variable on Membrane Permeability (Beetroot)
5. RP5: Dissection of Gas Exchange / Mass Transport System
6. RP6: Aseptic Technique & Antimicrobial Substances
7. RP7: Chromatography of Leaf Pigments
8. RP8: Effect of a Variable on Respiration Rate (Respirometer)
9. RP9: Choice Chamber (Animal Responses)
10. RP10: Distribution of Organisms (Quadrats & Transects)
11. RP11: Effect of a Variable on Organism Development
12. RP12: Immobilised Enzyme Investigation

© RAG Learning · Not for redistribution

RP1: Effect of a Variable on Enzyme-Controlled Reaction Rate

Aim: Investigate the effect of temperature (or pH, or substrate/enzyme concentration) on the rate of an enzyme-controlled reaction.

Biology link: Enzymes are biological catalysts (Topic 1). Their rate of reaction is affected by temperature (kinetic energy and denaturation), pH (ionisation of R groups in active site), and substrate concentration (collision frequency and saturation).

Apparatus

- Test tubes and rack, boiling tubes
- Catalase source (potato extract or liver homogenate) and hydrogen peroxide (H_2O_2) solution (typically 3-10 volume)
- Gas syringe (most accurate) or inverted measuring cylinder over water
- Delivery tube with bung (tight-fitting to prevent gas escape)
- Water baths at range of temperatures (e.g. 20, 30, 40, 50, 60°C) or buffer solutions for pH investigation
- Thermometer, stopwatch, graduated syringes or pipettes for measuring volumes

Method

1. Set up water bath at the required temperature. Place both enzyme and substrate solutions in separate test tubes in the water bath for **5 minutes** to equilibrate to the target temperature (this is a key step often missed).
2. Measure a fixed volume of enzyme and a fixed volume of H_2O_2 using graduated syringes.
3. Add the enzyme to the H_2O_2, immediately fit the bung and delivery tube connected to the gas syringe (or inverted measuring cylinder).
4. Record the volume of O_2 gas collected every 30 seconds for 5 minutes.
5. Repeat at each temperature (or pH/concentration). Perform **at least 3 repeats** per condition and calculate mean values.
6. Calculate the **initial rate of reaction** from the steepest (linear) part of the volume-time curve (typically the first 30-60 seconds).

© RAG Learning · Not for redistribution

Alternative method (for amylase): Amylase breaks down starch. Mix amylase with starch solution at different temperatures. At regular intervals, take a sample and add to iodine solution. Time how long until iodine no longer turns blue-black (starch fully digested). Rate = 1/time.

Variables

Independent	Dependent	Control Variables
Temperature (or pH, or concentration)	Volume of O_2 produced per unit time (rate of reaction)	Volume and concentration of enzyme, volume and concentration of substrate, same enzyme batch, same total volume, equilibration time

Safety

- H_2O_2 is an irritant at higher concentrations — wear eye protection and gloves.
- Care with hot water baths — do not exceed 60°C; handle tubes with tongs.
- O_2 is released — ensure good ventilation. Do not approach with naked flames.
- If using buffer solutions, some are irritant — check MSDS.

Analysis

- Plot volume of O_2 (y-axis) against time (x-axis) for each condition — gives a product-time curve.
- Calculate **initial rate** = gradient of the tangent at time = 0 (or from the linear portion). Rate = Δvolume / Δtime ($cm^3\ s^{-1}$ or $cm^3\ min^{-1}$).
- Plot initial rate (y-axis) against temperature or pH (x-axis) to find the **optimum**.
- For temperature: rate increases with temperature (more kinetic energy, more enzyme-substrate collisions) up to an optimum, then decreases sharply as the enzyme denatures (tertiary structure disrupted, active site shape changed permanently).
- For pH: bell-shaped curve with a peak at the optimum pH. Extreme pH values disrupt ionic bonds and hydrogen bonds in the enzyme, altering the active site shape.

Sources of Error and Improvements

Error	Effect	Improvement

Delay fitting bung after mixing	Gas escapes before collection begins; rate appears lower	Practise technique; use a side-arm flask with a rubber bung already in place
Temperature fluctuation during reaction	Rate varies inconsistently; enzyme may partially denature	Use a thermostatically controlled water bath; check temperature with thermometer
Enzyme concentration varies between batches	Unreliable comparisons between conditions	Use the same batch of enzyme preparation throughout; make one large batch and divide
Bubbles in delivery tube or syringe	Volume readings are inaccurate (artificially inflated)	Ensure delivery tube is filled with water before starting; check for leaks
Not equilibrating enzyme and substrate at target temperature	Reaction starts at wrong temperature; results invalid	Allow 5 minutes for both solutions to reach target temperature before mixing
Substrate depletion at end of reaction	Rate slows as substrate runs out, not due to variable being tested	Use initial rate (first 30-60 s) when substrate is in excess

Exam Tip: Always state that you would calculate the *initial rate* of reaction (not just "rate"). The rate at time = 0 avoids issues with substrate depletion, product inhibition, and enzyme inactivation. To calculate it: draw a tangent to the curve at t = 0 (or use the first linear portion) and measure its gradient. Also: when discussing temperature, distinguish between increased kinetic energy (reversible) and denaturation (irreversible — bonds broken, active site permanently changed).

RP2: Stained Squashes of Root Tip Cells (Mitosis)

Aim: Observe mitosis in root tip cells and calculate the mitotic index.

Biology link: The root tip meristem contains actively dividing cells (Topic 2). The mitotic index can indicate how rapidly a tissue is growing and has clinical relevance — a high mitotic index in tumour tissue indicates aggressive cancer.

Apparatus

- Garlic or onion root tips (grown in water for 3-5 days; garlic is often preferred as it has large chromosomes and a fast growth rate)
- Acetic orcein stain (or toluidine blue / Feulgen stain) — stains chromosomes dark red/purple
- 1 M hydrochloric acid, watch glass, water bath (or Bunsen burner with care)
- Microscope slides, coverslips, filter paper
- Optical microscope (x100 and x400 magnification), mounted needle, pencil eraser or thumb

Method

1. Cut approximately 2 mm of the **growing root tip** (the meristematic region where cells are actively dividing) and place on a watch glass.
2. Add a few drops of 1 M HCl and warm gently at ~60°C for 5 minutes in a water bath. This **hydrolyses the middle lamella** (breaks down pectin between cell walls), softening the tissue and separating cells so they form a single layer when squashed.
3. Carefully blot dry with filter paper, then add a few drops of **acetic orcein stain**. Leave for **10-15 minutes**. The stain binds to DNA, making chromosomes visible (darkly stained).
4. Transfer to a clean microscope slide. Add a coverslip. Place filter paper over the coverslip and **squash firmly** with a pencil eraser (press vertically, do not twist — this would smear cells). This spreads cells into a single layer.
5. View under microscope: start at **low power (x100)** to locate a good region of well-spread cells, then switch to **high power (x400)** to identify individual cells.
6. Identify and count cells in each stage of mitosis (prophase, metaphase, anaphase, telophase) and those in interphase. Count at least **200 cells** from multiple fields of view for reliability.

Safety

- HCl is corrosive — wear gloves and eye protection. Handle hot HCl with extra care.

© RAG Learning · Not for redistribution

- Acetic orcein stains skin and clothing permanently — handle carefully, wear gloves and lab coat.
- Care with hot water or Bunsen burner. Water bath is safer than direct heating.
- Sharp instruments (scalpels, needles) — cut away from body on a tile.

Analysis

$$\text{Mitotic index} = \text{number of cells undergoing mitosis} \,/\, \text{total number of cells observed}$$

- Record the number of cells at each stage: interphase, prophase, metaphase, anaphase, telophase.
- The **proportion of cells at each stage reflects the relative duration of that phase**. Prophase is the longest phase of mitosis, so most dividing cells will be in prophase. Anaphase is the shortest, so fewest cells will be seen in this stage.
- A higher mitotic index indicates faster cell division (more cells in mitosis at any one time).
- Clinically, mitotic index is used to assess **tumour aggressiveness** — higher values indicate faster growth.

Sources of Error and Improvements

Error	Effect	Improvement
Cells overlapping after squash	Difficult to identify individual cells and stages	Squash more firmly (but do not twist); use thinner tip sections; ensure adequate HCl treatment
Insufficient staining time	Chromosomes not visible; difficult to identify stages	Leave stain for at least 10-15 minutes; ensure fresh stain is used
Counting non-meristematic cells	Lower mitotic index (mature cells do not divide)	Only cut and use the very tip (2 mm) where the meristem is located
Small sample size	Unreliable and unrepresentative mitotic index	Count at least 200 cells from multiple fields of view across the slide

© RAG Learning · Not for redistribution

Difficulty distinguishing phases	Misidentification of stages	Use reference images; practise identification; use higher magnification
Air bubbles under coverslip	Distort image; obscure cells	Lower coverslip at an angle; add more stain if needed

Exam Tip: Prophase typically has the most cells (longest phase), anaphase the fewest (shortest). Always state the formula: *mitotic index = cells in mitosis / total cells observed*. If asked to identify stages from a photograph: prophase = chromosomes visible but scattered; metaphase = chromosomes aligned at equator; anaphase = chromosomes being pulled apart to poles (V-shaped); telophase = two groups of chromosomes at poles, cell beginning to divide. Interphase cells show a dark nucleus with no visible individual chromosomes.

RP3: Dilution Series & Calibration Curve (Colorimetry)

Aim: Produce a calibration curve using a serial dilution and colorimetry, then use it to determine the concentration of an unknown solution.

Biology link: Quantitative analysis of biological molecules (Topic 1). Colorimetry is used to measure concentrations of coloured solutions — applicable to enzyme assays, glucose testing (Benedict's), protein quantification (biuret), and many medical diagnostic tests.

Apparatus

- Stock solution of known concentration (e.g. glucose solution, food dye, or bovine serum albumin)
- Distilled water, graduated pipettes or syringes (accurate to 0.1 cm^3)
- Test tubes and rack (at least 6 — for serial dilution + blank + unknown)
- Colorimeter (or spectrophotometer) and clean cuvettes

- Appropriate colour filter for the colorimeter (use the **complementary colour** to the solution — e.g. blue/green filter for a red/orange solution like Benedict's)
- Reagent for colour development if needed (e.g. Benedict's reagent for reducing sugars, biuret reagent for protein)

Method

1. **Prepare a serial dilution:** Transfer a fixed volume (e.g. 5 cm^3) of stock solution to the first tube. Add an equal volume (5 cm^3) of distilled water and mix thoroughly. Transfer 5 cm^3 from this to the next tube. Add 5 cm^3 distilled water. Repeat for 5-6 tubes total. This gives a doubling dilution series (100%, 50%, 25%, 12.5%, 6.25%, 3.125%).
2. If a colour-development reagent is needed, add the same volume to each tube (including a blank of distilled water + reagent). Allow colour to develop fully (e.g. heat Benedict's in a water bath at 80°C for 5 minutes).
3. **Zero/calibrate the colorimeter** using the blank (distilled water + reagent, no analyte).
4. Measure the **absorbance** (or % transmission) of each known concentration solution in the colorimeter. Use clean, fingerprint-free cuvettes oriented consistently.
5. Plot **absorbance (y-axis) against concentration (x-axis)** to produce a **calibration curve**.
6. Measure the absorbance of the **unknown solution** (treated identically) and read off its concentration from the calibration curve.

Safety

- Benedict's reagent contains copper sulfate — irritant. Wear eye protection.
- Biuret reagent contains NaOH — corrosive. Handle with care, wear gloves.
- Hot water baths — use tongs; do not overfill tubes.

Analysis

- Draw a **line of best fit** through the calibration points (should be approximately linear at lower concentrations — Beer-Lambert law region).
- Read the unknown concentration: draw a **horizontal line from its absorbance value to the calibration curve**, then a **vertical line down to the x-axis**. Read the concentration.

© RAG Learning · Not for redistribution

- If the unknown's absorbance falls outside the calibration range, it must be diluted and re-measured (then multiply by the dilution factor).
- At high concentrations, the calibration curve may plateau (deviates from linearity) — only interpolate within the linear range.

Sources of Error and Improvements

Error	Effect	Improvement
Air bubbles in cuvette	Scatter light; give incorrectly high absorbance readings	Tap cuvette gently to dislodge bubbles; wipe with lens tissue
Inaccurate pipetting during dilution	Dilution concentrations wrong; calibration curve unreliable	Use calibrated graduated pipettes; pipette at eye level; practise technique
Fingerprints or scratches on cuvette	Light absorbed/scattered by contaminants	Handle cuvettes by the top edges only; wipe clear sides with lens tissue; always orient cuvette the same way
Wrong colour filter selected	Low sensitivity — small differences between concentrations not detected	Use the **complementary colour** filter (the colour that the solution absorbs most — i.e. the opposite colour on the colour wheel)
Not mixing dilutions thoroughly	Concentration gradient within tube; inaccurate readings	Invert tube 3-4 times or vortex after each dilution step

Exam Tip: A serial dilution gives a wide range of concentrations in equal ratio steps. Each 1-in-2 dilution halves the concentration. Always use the *same volume* transferred at each step and the *same volume of diluent*. The colorimeter measures absorbance (how much light is absorbed by the solution) — higher concentration = darker colour = higher absorbance.

> Make sure you understand when to use absorbance vs % transmission (absorbance is proportional to concentration; % transmission is inversely related).

RP4: Effect of a Variable on Membrane Permeability (Beetroot)

Aim: Investigate the effect of temperature (or alcohol concentration, or pH) on cell membrane permeability, using beetroot as a model system.

Biology link: Cell membrane structure (Topic 2). Beetroot cells contain the red-purple pigment **anthocyanin** in their vacuoles. The tonoplast (vacuolar membrane) and cell surface membrane normally retain this pigment inside the cell. When membranes are disrupted, anthocyanin leaks out into the surrounding solution. The amount of leakage (measured by colorimetry) indicates the degree of membrane damage.

Apparatus

- Fresh beetroot, cork borer (standardised size), white tile, sharp knife/scalpel
- Test tubes and rack, distilled water (measured volumes, e.g. 5 cm^3)
- Water baths at range of temperatures (e.g. 20, 30, 40, 50, 60, 70°C)
- Colorimeter and cuvettes (using green or blue filter — complementary to red pigment)
- Thermometer, stopwatch, paper towels

Method

1. Use a cork borer to cut equal-sized beetroot cylinders (or use a ruler and knife to cut uniform discs of equal diameter and thickness). This standardises the **surface area** exposed.
2. **Rinse thoroughly** in cold distilled water until the water runs clear. This removes any anthocyanin released during cutting from damaged surface cells. This is critical — without adequate rinsing, all samples will appear to have high leakage.
3. Place one disc in each labelled test tube containing **5 cm^3 distilled water** (measured precisely).

4. Place each tube in the appropriate water bath for a **standardised time** (e.g. exactly 20 minutes).
5. After exactly 20 minutes, remove the beetroot discs. Gently agitate each solution to distribute the pigment evenly.
6. Measure the **absorbance** of each solution using the colorimeter with a green/blue filter (zeroed with distilled water blank).
7. Repeat **3 times at each temperature** and calculate the mean absorbance.

Safety

- Sharp instruments (scalpel, cork borer) — cut on a white tile, away from the body.
- Hot water baths — handle tubes with tongs; take care with boiling water.
- Beetroot stains clothing and skin — wear lab coat and gloves if concerned.

Expected Results and Explanation

- At **low temperatures (20-30°C)**: low absorbance — membranes are intact, little pigment leaks out.
- At **moderate temperatures (30-40°C)**: slight increase — phospholipid bilayer becomes slightly more fluid, minor increase in permeability.
- At **high temperatures (above ~40°C)**: sharp increase in absorbance — membrane proteins **denature** (tertiary structure disrupted, proteins unfold and leave gaps in the membrane) AND phospholipids become much more fluid, increasing membrane permeability dramatically. Anthocyanin leaks freely.
- At very high temperatures (60-70°C): maximum leakage — membranes essentially destroyed.

Sources of Error and Improvements

Error	Effect	Improvement
Unequal disc sizes	Variable surface area affects amount of pigment released	Use same cork borer; cut all discs to uniform thickness with a ruler
Insufficient rinsing of cut surfaces	Surface pigment from damaged cells inflates all readings	Rinse until water runs completely clear before starting

© RAG Learning · Not for redistribution

Different time in water bath	Longer exposure = more leakage regardless of temperature	Use a stopwatch; stagger start times at 1-minute intervals; time precisely to 20 minutes each
Cells damaged during cutting	Extra pigment released from damaged cells (not due to temperature)	Discard end pieces; use only middle sections; handle gently; use a sharp borer
Pigment settling in tube	Uneven colour distribution; inconsistent readings	Agitate solution gently before transferring to cuvette

Exam Tip: Link results directly to membrane structure. At high temperatures: (1) **phospholipids gain kinetic energy** and move more, increasing membrane fluidity and creating gaps; (2) **membrane proteins denature** — hydrogen bonds and ionic bonds within the protein's tertiary structure are broken, the protein unfolds and can no longer maintain the membrane's barrier function. Mention both protein denaturation AND phospholipid fluidity for full marks. If asked about alcohol: ethanol dissolves phospholipids (it is a solvent for lipids), disrupting the bilayer structure and increasing permeability.

RP5: Dissection of Gas Exchange / Mass Transport System

Aim: Dissect an organism or organ to examine adaptations for gas exchange or mass transport (e.g. fish head/gills, heart, lungs, leaf).

Biology link: Gas exchange surfaces (Topic 3). Organisms have evolved specialised surfaces with features that maximise the rate of diffusion: large surface area, thin barrier (short diffusion distance), maintained concentration gradient (ventilation, blood supply).

Apparatus

- Fresh fish head (from fishmonger — not preserved), OR heart (lamb/pig), OR lungs
- Dissecting board, pins, scissors, forceps, scalpel, mounted needle

- Magnifying glass or dissecting microscope
- Paper towels, gloves, eye protection, lab coat

Method (Fish Gills)

1. Place the fish head on the dissecting board. Lift the **operculum** (gill cover) to expose the gills beneath.
2. Use scissors to carefully cut away the operculum, exposing the gill arches.
3. Identify the **4 gill arches** on each side, the **gill filaments** (primary lamellae — feathery pink structures), and the **gill rakers** (bony projections that filter food particles from water).
4. Remove one gill arch and place it in a dish of water on a white tile. Use a magnifying glass or dissecting microscope to observe the **secondary lamellae** (thin plate-like projections from each filament — the actual gas exchange surfaces).
5. Produce a **labelled biological drawing**: use clear, continuous lines; no shading; labels with ruled lines; include a title, magnification, and scale.

Safety

- Wear gloves when handling biological specimens (bacterial contamination risk).
- Sharp instruments — always cut away from the body on a dissecting board.
- Wash hands thoroughly with antibacterial soap after the practical.
- Dispose of biological material in designated biological waste bins (not regular bins).

Analysis — Structure-Function Relationships

Adaptation	How It Maximises Gas Exchange
Many gill filaments and lamellae	Provides a very **large surface area** for diffusion of O_2 and CO_2
Lamellae are very thin (one cell thick epithelium)	**Short diffusion distance** — gases cross quickly by diffusion
Dense network of capillaries in each lamella	Rich blood supply maintains a steep **concentration gradient**; carries O_2 away and brings CO_2

Countercurrent flow (water and blood flow in opposite directions)	Maintains a concentration gradient along the **entire length** of the lamella — achieving ~80% O_2 extraction (much more efficient than parallel flow)
Ventilation mechanism (buccal pumping)	Continuous flow of fresh water over gills maintains the concentration gradient

Exam Tip: In dissection questions, always link **structure to function**. For any gas exchange surface, mention: (1) large surface area, (2) thin barrier/short diffusion distance, (3) mechanism to maintain concentration gradient (ventilation + blood supply). For fish specifically, explain **countercurrent flow**: because blood flows in the opposite direction to water, there is always a diffusion gradient at every point along the lamella — this is far more efficient than if they flowed in the same direction (parallel flow), where equilibrium would be reached partway along.

RP6: Aseptic Technique & Antimicrobial Substances

Aim: Use aseptic technique to investigate the effect of antimicrobial substances (antibiotics or plant extracts) on bacterial growth.

Biology link: Infectious disease and antibiotics (Topic 3/6). Antibiotic resistance is a major global health concern. This practical demonstrates how antibiotics inhibit bacterial growth and how their effectiveness can be compared quantitatively.

Apparatus

- Nutrient agar plates (pre-poured and sterilised by autoclaving)
- Broth culture of bacteria (e.g. non-pathogenic *Escherichia coli* or *Micrococcus luteus*)
- Antibiotic discs (multidiscs or individual, e.g. penicillin, streptomycin, tetracycline) and/or plant extract soaked filter paper discs (e.g. garlic, mint, tea tree oil)
- Sterile inoculating loop or sterile spreader, Bunsen burner

- Forceps (to be flamed), marker pen (for labelling base of plate), adhesive tape, ruler, calculator

Aseptic Technique — Key Steps

- Work near a **lit Bunsen burner** — the updraft creates a convection current that carries airborne microorganisms upward, away from the work area (creating an **aseptic zone**).
- **Flame the inoculating loop** until red hot before and after use — this sterilises it by killing all microorganisms.
- **Flame the neck of culture bottles** briefly after opening and before closing — kills microorganisms at the rim.
- **Do not breathe on** or cough over open plates or cultures.
- Keep lids on plates as much as possible; only open when necessary.
- Wipe down the bench with disinfectant before and after the practical.

Method

1. Flame the inoculating loop, allow to cool briefly, dip into bacterial broth culture, and **spread evenly** across the agar plate surface using a zigzag pattern (streak plate method) or use a sterile glass spreader.
2. Using **flamed forceps**, place antibiotic discs (or plant extract discs) at equal distances on the agar surface. Press gently to ensure contact with the agar.
3. **Seal the plate** with two strips of adhesive tape in a cross pattern (do NOT seal completely around the edge — this could create **anaerobic conditions** inside the plate, promoting the growth of dangerous anaerobic pathogens).
4. Label the **base** of the plate (not the lid — lids can be accidentally swapped) with name, date, and disc identities.
5. Incubate the plate **upside down** (prevents condensation dripping onto colonies, which could spread bacteria and blur zones) at a **maximum of 25°C** for 24-48 hours.
6. After incubation, measure the **diameter of each clear zone of inhibition** (the area around each disc where bacteria have not grown) using a ruler. Do **NOT open the plate**.

Safety

- Incubate at **≤25°C in school laboratories** (not at 37°C — human body temperature would encourage growth of human pathogens that may have contaminated the plate).

- **Do not open plates after incubation** — unknown organisms may have grown.
- **Autoclave all plates** (121°C, 15 min) before disposal to kill all microorganisms.
- Maintain strict aseptic technique throughout to prevent contamination of the plate AND to prevent release of microorganisms into the laboratory.
- Wash hands with antibacterial soap before and after the practical.

Analysis

- Measure the **diameter** of each zone of inhibition (in mm). Take two measurements at right angles and average.
- Calculate the **area** of each zone: $A = \pi r^2$ (where r = diameter / 2). **Always use area, not diameter, for comparison** — area is proportional to the amount of antibiotic that has diffused, while diameter is not (area increases with the square of the radius).
- Larger zone of inhibition = more effective antimicrobial substance (inhibits bacterial growth over a wider area).
- Subtract the area of the disc itself from the total zone area for a more accurate comparison.

Sources of Error and Improvements

Error	Effect	Improvement
Contamination from air	Unwanted colonies of environmental microorganisms grow on the plate	Work near a Bunsen burner; flame all equipment; minimise time plate lid is off
Uneven bacterial spread	Zones of inhibition are irregular or misleading	Use a sterile spreader with a turntable; rotate plate 90° between streaks
Discs placed too close together	Overlapping zones; cannot measure accurately	Space discs evenly (at least 2 cm apart); use a template or multipoint inoculator
Comparing diameters instead of areas	Underestimates differences in effectiveness (area scales with r^2)	Always calculate πr^2 for valid quantitative comparison

Inconsistent disc contact with agar	Poor diffusion; smaller zones than expected	Gently press each disc onto agar surface with sterile forceps to ensure good contact

Exam Tip: Two key points examiners look for: (1) Always calculate the **area** of the zone of inhibition (πr^2), not just the diameter — marks are specifically allocated for this. (2) Know **why** plates are incubated at ≤25°C in schools — to reduce the risk of growing human pathogens (which thrive at 37°C). In industrial/hospital labs, plates ARE incubated at 37°C because facilities are appropriate for handling pathogens safely.

RP7: Chromatography of Leaf Pigments

Aim: Separate and identify photosynthetic pigments from a leaf extract using thin-layer or paper chromatography.

Biology link: Photosynthesis (Topic 5). Chloroplasts contain multiple pigments that absorb different wavelengths of light — this broadens the range of light energy that can be captured. Separating them demonstrates that leaves contain a mixture of pigments, not just chlorophyll.

Apparatus

- Fresh green leaves (spinach, nettle, or grass), pestle and mortar, clean sand (abrasive), small volume of propanone (acetone)
- Chromatography paper (or TLC plate with silica gel), pencil (NOT pen), ruler
- Boiling tube or chromatography jar with tight-fitting lid, capillary tube or micropipette
- Running solvent: propanone (acetone) or petroleum ether : acetone (9:1) mix

Method

1. Grind leaves with a small amount of clean sand (to break open cells and chloroplasts) and a few cm^3 of propanone in a mortar. The sand acts as an abrasive. Filter the deep green extract through muslin or filter paper.

2. Draw a **pencil line** (origin line) 2 cm from the bottom of the chromatography paper. Use pencil because ink would dissolve in the solvent and interfere with results.
3. Apply leaf extract to the origin line using a capillary tube — build up a **concentrated spot** by applying small amounts repeatedly, allowing to dry between applications (5-10 applications).
4. Place the paper in the jar with the solvent level **below the origin line** (so the spot is above the solvent — otherwise pigments would dissolve directly into the solvent pool rather than being carried up by capillary action). Seal the jar with a lid to prevent solvent evaporation.
5. Allow the solvent to rise by capillary action until it is approximately 1-2 cm from the top. **Mark the solvent front** immediately with pencil before it evaporates.
6. Remove the paper and allow it to dry. Measure the distance from the origin line to the centre of each pigment spot, and the distance from the origin to the solvent front.

Safety

- Propanone (acetone) is **highly flammable** and produces irritating fumes — NO naked flames near the experiment; work in a well-ventilated area or fume cupboard.
- Petroleum ether is extremely flammable — same precautions.
- Avoid prolonged skin contact with solvents — wear nitrile gloves.

Analysis

$$R_f = \text{distance moved by pigment from origin} / \text{distance moved by solvent front from origin}$$

- Calculate R_f value for each separated pigment. Compare to reference/published R_f values to **identify each pigment**.
- Typical order of separation (highest R_f = most soluble in the non-polar solvent, travels furthest):

Pigment	Colour	Approximate R_f	Solubility
Carotene	Orange-yellow	~0.95	Most soluble (non-polar, dissolves readily in non-polar solvent)

Xanthophyll	Yellow	~0.7	Moderately soluble
Chlorophyll a	Blue-green	~0.5	Less soluble
Chlorophyll b	Yellow-green	~0.4	Least soluble (more polar; held more strongly by the stationary phase)

Sources of Error and Improvements

Error	Effect	Improvement
Origin line below solvent level	Pigments dissolve directly into solvent and are washed away	Ensure the solvent level is 1 cm below the origin line
Jar lid not sealed properly	Solvent evaporates; uneven solvent front; inconsistent R_f values	Use a tight-fitting lid or cover with cling film; line jar with filter paper soaked in solvent to saturate the atmosphere
Extract not concentrated enough	Spots too faint to see or measure	Apply many small applications, drying between each; use a concentrated extract
Using pen instead of pencil for origin	Pen ink dissolves in solvent and runs up the paper, contaminating results	Always use a pencil line
Not marking solvent front immediately	Solvent evaporates; R_f values cannot be calculated accurately	Mark the solvent front with pencil the instant you remove the paper

Exam Tip: Carotene has the highest R_f because it is the most non-polar pigment and therefore most soluble in the non-polar organic solvent. Chlorophyll b has the lowest R_f because it is the most polar (it has more polar -CHO groups compared to chlorophyll a's $-CH_3$

group) and is more attracted to the polar stationary phase (paper/silica). R_f values are specific to a given solvent system and temperature — always state the solvent used.

RP8: Effect of a Variable on Respiration Rate (Respirometer)

Aim: Investigate the effect of a variable (temperature, type of substrate, or organism) on the rate of aerobic respiration, using a respirometer to measure O_2 consumption.

Biology link: Respiration (Topic 5). All living organisms respire continuously. The rate of O_2 consumption is a direct measure of aerobic respiration rate. Temperature affects enzyme activity (respiratory enzymes in glycolysis, Krebs cycle, and ETC).

Apparatus

- Simple respirometer (two tubes connected to a capillary tube/manometer with a scale)
- Organisms: germinating seeds (e.g. peas, mung beans), small invertebrates (e.g. woodlice, blowfly larvae), or yeast suspension
- Soda lime or potassium hydroxide pellets (to absorb CO_2 produced by respiration)
- Capillary tube with graduated scale, coloured oil or fluid drop (indicator of gas volume change)
- Water bath (for temperature control), thermometer, stopwatch
- Glass beads (for the control tube — same volume as organisms, non-living)

How the Respirometer Works

The key principle: soda lime (or KOH) **absorbs all CO_2** produced by the organisms. Therefore, the only gas change is the **consumption of O_2**. As O_2 is used up, the gas volume in the sealed tube decreases, and the **coloured fluid drop moves towards the organisms**. The distance moved per unit time = rate of O_2 uptake = rate of aerobic respiration.

The **control tube** (containing glass beads + soda lime, but no living organisms) accounts for any changes in gas volume due to temperature or atmospheric pressure changes that are NOT due to respiration.

Method

1. Set up the respirometer: place organisms (e.g. germinating seeds) in one tube with a layer of soda lime at the base (separated from organisms by cotton wool to prevent direct contact). Set up a **control tube** with glass beads of equal volume and soda lime.
2. Seal both tubes and connect them to the manometer/capillary tube via airtight connections (use Vaseline on joints to prevent leaks).
3. Place in a water bath at the desired temperature. Allow **5 minutes to equilibrate** (temperature stabilisation and organisms to settle).
4. Introduce a coloured fluid drop (or oil droplet) into the capillary tube using a syringe.
5. Record the **distance the fluid drop moves** towards the organisms every minute for 10-15 minutes.
6. Repeat at different temperatures (e.g. 15, 20, 25, 30, 35, 40°C). Calculate rate of O_2 uptake (mm min^{-1} or mm^3 min^{-1} if capillary tube cross-section is known).

Safety

- Soda lime and KOH are **corrosive** — wear gloves and eye protection. Avoid direct contact with organisms.
- Handle organisms **ethically** — return woodlice to their habitat after the experiment; do not expose to extreme temperatures.
- Do not heat organisms above 40°C or below 5°C.

Analysis

- **Rate of O_2 consumption** = distance moved by fluid drop / time (mm min^{-1}).
- Plot rate of O_2 uptake (y-axis) against temperature (x-axis). Expect a general increase with temperature (Q_{10} effect on enzyme activity) up to an optimum, then a decrease at high temperatures (enzyme denaturation).
- The **control tube** corrects for any volume changes due to atmospheric pressure and temperature fluctuations. If the control drop moves, subtract this from the experimental reading.

Sources of Error and Improvements

Error	Effect	Improvement
Leaks in apparatus	Air enters; fluid drop does not move correctly (or moves the wrong way)	Check all connections; apply Vaseline to all joints; test for leaks before starting
Soda lime exhausted or insufficient	CO_2 not fully absorbed; gas volume change is less than expected (CO_2 partially replaces O_2)	Use fresh, excess soda lime for each experiment
Temperature fluctuation in water bath	Affects gas expansion/contraction independently of respiration	Use a thermostatically controlled water bath; allow adequate equilibration time
Organisms stressed by handling	Abnormally high or low respiration rate initially	Allow a 5-minute equilibration period after setup before starting measurements
Dead organisms in sample	Lower apparent respiration rate	Use freshly germinated seeds or active organisms; check for viability

Exam Tip: The soda lime absorbs CO_2, so **only O_2 consumption** causes the gas volume change. Without soda lime, the CO_2 produced would partially replace the O_2 consumed, and the volume change would be much smaller (depending on the RQ of the substrate). The control tube is essential — always explain its purpose: it accounts for any volume changes due to factors other than respiration (atmospheric pressure changes, temperature fluctuations). To measure anaerobic respiration, you would remove the soda lime and measure CO_2 production instead.

RP9: Choice Chamber (Animal Responses to Stimuli)

Aim: Investigate the effect of an environmental factor (light/dark, humidity) on the distribution and movement of small invertebrates (e.g. woodlice, maggots).

Biology link: Organisms respond to stimuli (Topic 6). Woodlice show **kinesis** (non-directional response — they change their speed or rate of turning in response to stimulus intensity) rather than taxis (directional movement towards or away from a stimulus). They tend to move slower and turn less in favourable conditions (dark, humid), accumulating there over time.

Apparatus

- Choice chamber (circular or rectangular, divided into 2-4 compartments with a perforated divider and clear lid)
- 10-20 woodlice (collected humanely from under logs, stones, or leaf litter)
- Damp cotton wool (for humid conditions) and dry silica gel / anhydrous calcium chloride (desiccant, for dry conditions)
- Black card or aluminium foil (for dark conditions), lamp (for light conditions)
- Stopwatch, data recording sheet

Method

1. Set up **two contrasting environments** in the choice chamber. Test **only one variable at a time**:
   - For humidity: one side has damp cotton wool (humid), the other has dry silica gel (dry) — both kept at the same light level
   - For light: one side is covered with black card (dark), the other is exposed to a lamp (light) — both kept at the same humidity
2. Place 10 woodlice in the **centre** of the chamber (on the divider), ensuring they have equal access to both sides.
3. Replace the lid and start the timer immediately.
4. Count the number of woodlice on each side **every 2 minutes for 20-30 minutes**. Record in a table.
5. Repeat the entire experiment **at least 3 times** with different woodlice. Calculate mean number on each side at each time interval.

Safety and Ethics

- Handle woodlice **gently** — they are living organisms deserving ethical treatment.
- **Return all woodlice** to their original habitat after the experiment.
- Do not expose to extreme conditions (very hot, direct strong light) for extended periods.
- Wash hands after handling. Silica gel can be irritating — avoid direct skin contact.

Analysis

- Calculate the **mean number** of woodlice on each side at each time interval across all repeats.
- Woodlice show **kinesis**: they move more slowly and turn less in preferred conditions (dark, humid), so they **accumulate** in the dark/humid side over time — not because they are attracted to it, but because they spend more time there once they arrive.
- Use a **chi-squared test** to determine if the final distribution is significantly different from the 50:50 distribution expected if woodlice showed no preference (null hypothesis: no significant difference from a 1:1 distribution).

Sources of Error and Improvements

Error	Effect	Improvement
Woodlice on the divider/boundary	Difficult to assign to one side	Decide a consistent rule beforehand (e.g. assign by head position); exclude borderline individuals from count
Two variables changed at once (e.g. dark AND humid on same side)	Cannot determine which factor causes the preference	Only vary ONE factor per experiment; keep all other conditions identical
Small sample size / few repeats	Results may not be statistically significant or representative	Use at least 10 woodlice per trial; repeat at least 3 times; pool data from class
Temperature differences between sides	Confounding variable — woodlice may respond to	Ensure both sides are at the same temperature (no lamp heat on

	temperature, not the intended variable	one side only — use a distant lamp or LED)
Organisms stressed by handling	Erratic initial behaviour unrelated to the variable	Allow a 2-minute settling period before starting to count

Exam Tip: Woodlice show **kinesis, not taxis**. Kinesis is a **non-directional** response where the organism changes its **speed of movement** or **rate of turning** depending on stimulus intensity. In unfavourable conditions (light, dry), they move faster and turn more frequently (increasing the chance of finding better conditions). In favourable conditions (dark, humid), they slow down and turn less — so they accumulate there. This is different from taxis, which is a directional movement towards or away from a stimulus (e.g. positive phototaxis in Euglena).

RP10: Distribution of Organisms (Quadrats & Transects)

Aim: Investigate the distribution of organisms in a habitat and the effect of an abiotic factor on their distribution, using appropriate sampling techniques.

Biology link: Ecology and biodiversity (Topics 4 and 7). The distribution of organisms is influenced by both abiotic factors (light, temperature, soil pH, moisture, exposure) and biotic factors (competition, predation). Sampling must be done appropriately to give a representative picture of the community.

Apparatus

- Quadrats (e.g. 0.5 m x 0.5 m frame, possibly subdivided into 25 smaller squares for % cover estimation)
- Tape measure (30 m or longer) for transect line
- Random number generator or random number table (for random quadrat placement)
- Species identification keys, recording sheets, pencil
- Abiotic measurement tools: light meter, soil pH probe/meter or BDH soil test kit, soil moisture meter, thermometer, anemometer (wind speed)

Two Sampling Approaches

Method	When to Use	How
Random quadrat sampling	Estimating the abundance/density of species in a **uniform habitat** (e.g. a field)	Use random number pairs as x,y coordinates to position each quadrat. This avoids sampling bias. Count organisms or estimate % cover in each quadrat. Repeat many times (at least 10-20).
Systematic sampling along a transect	Studying **changes in species distribution along an environmental gradient** (e.g. from seashore to cliff top, from woodland edge to interior, from path to field centre)	Lay a tape measure along the gradient. Place quadrats at regular intervals (e.g. every 2 metres). Record species and abundance + abiotic factors at each position.

Method (Belt Transect Example)

1. Lay a tape measure in a straight line across the habitat, along the environmental gradient of interest (e.g. from the high water mark up a rocky shore).
2. Place a quadrat at regular intervals along the transect (e.g. every 2 metres, or every 5 metres for longer transects).
3. At each quadrat position:
   - Identify and record all species present (use identification keys)
   - Estimate % cover for each plant species (or count individuals for animals)
   - Measure relevant abiotic factors at the same point (light intensity, soil pH, soil moisture, temperature, wind speed)
4. Record all data in a prepared table. Repeat the transect in parallel (if possible) for reliability.

Safety

- Conduct a **risk assessment** for fieldwork: assess terrain (uneven ground, cliffs, slippery rocks), weather conditions, tidal risks (on a seashore), and lone working policy.

- Work in pairs or groups. Carry a mobile phone. Inform someone of your location and expected return time.
- Wash hands after handling soil (may contain tetanus bacteria, Weil's disease from rat urine).
- Be aware of hazardous plants (stinging nettles, thorns, giant hogweed) and animals (ticks, adders).
- Wear appropriate clothing and footwear for the habitat.

Analysis

- Calculate **species density** = mean count per quadrat / quadrat area (number per m^2).
- Calculate **% frequency** = (number of quadrats containing the species / total number of quadrats) x 100.
- Use a **kite diagram** or bar chart to display species distribution along a transect.
- Calculate **Simpson's Diversity Index**: $D = 1 - \Sigma(n/N)^2$.
- Use **Spearman's rank correlation** to test for a significant correlation between species abundance and an abiotic factor (e.g. "Is there a significant correlation between % cover of a plant species and light intensity?").
- Spearman's r_s ranges from -1 (perfect negative correlation) through 0 (no correlation) to +1 (perfect positive correlation). Compare r_s to the critical value for your sample size at $p = 0.05$.

Sources of Error and Improvements

Error	Effect	Improvement
Non-random placement of quadrats (e.g. choosing "interesting" spots)	Sampling bias — results not representative of the whole habitat	Use a random number table/generator for coordinates; place quadrats without looking
Too few quadrats	Unrepresentative data; poor statistical power	Use at least 10-20 quadrats per area; pool class data
Misidentification of species	Inaccurate species count and diversity calculation	Use reliable identification keys; take photographs for verification; have an expert check

Seasonal/time-of-day variation	Different results at different times	Sample at the same time of day and same season for valid comparisons
Trampling or disturbing the habitat	Alters the community being studied	Approach carefully; use a random walk pattern; minimise time in each area

Exam Tip: Random sampling avoids bias — always explain how you would achieve randomness (random number coordinates from a baseline and side-line). For **transects**, explain why systematic sampling is appropriate: you want to detect changes in distribution along a gradient, which random sampling might miss. Know when to use each statistical test: **chi-squared** for comparing observed vs expected frequencies (categories); **Spearman's rank correlation** for testing the relationship between two continuous variables (e.g. species abundance vs distance along transect); **Student's t-test** for comparing means of two groups.

RP11: Effect of a Variable on Organism Development

Aim: Investigate the effect of an environmental variable (light intensity, temperature, water availability, mineral ions) on the growth and development of plants (seed germination and seedling growth).

Biology link: Plant responses and growth (Topics 5/6). Light, temperature, water, and mineral ions all affect plant growth. This practical also tests understanding of experimental design, control of variables, and statistical analysis.

Apparatus

- Seeds (e.g. cress, mustard, or mung bean — fast-growing species are best)
- Petri dishes or seed trays, cotton wool or vermiculite, distilled water
- Light source (lamp at different distances or different wattages), or dark cupboard, or coloured filters

- Ruler (mm), labels, marker pen, thermometer
- Incubator or windowsill locations at different temperatures (if investigating temperature)

Method

1. Place equal numbers of seeds (e.g. 10-20) on damp cotton wool in each labelled Petri dish.
2. Set up different conditions for the independent variable being tested (e.g. different light intensities by varying distance from lamp, or different temperatures, or different concentrations of mineral solution).
3. Keep **all other variables constant**: same number of seeds per dish, same volume of water added, same temperature (if not the variable), same seed batch, same type of cotton wool.
4. Record observations daily for 10-14 days:
   - **Germination percentage** = (number germinated / total seeds) x 100
   - **Shoot height** (mm) — measure from base to tip of tallest shoot
   - Number of leaves, root length, fresh/dry mass (if appropriate)
5. Set up **at least 3 replicates** for each condition and calculate mean values.

Safety

- Some seeds may be treated with fungicide — wash hands after handling.
- Some seeds are toxic if ingested — do not eat any experimental material.
- Avoid inhalation of vermiculite dust if used as a growing medium.
- Mould may grow on damp cotton wool — dispose of mouldy dishes appropriately.

Analysis

- Calculate **germination percentage** at each time point for each condition.
- Plot **mean shoot height** (y-axis) against **time** (x-axis) for each condition on the same axes for comparison.
- Calculate **mean and standard deviation** for each condition — standard deviation indicates the spread/variability of the data. Plot error bars on graphs.
- Use a **Student's t-test** (for two conditions) or **one-way ANOVA** (for multiple conditions) to determine if differences in mean shoot height between conditions are statistically significant.

- Alternatively, use the **Mann-Whitney U test** if data are not normally distributed.

Sources of Error and Improvements

Error	Effect	Improvement
Variable seed viability	Some seeds may not germinate regardless of conditions	Use seeds from the same packet/batch; pre-test a sample for viability
Unequal watering	Confounding variable — water availability affects growth independently	Measure exact volume of water added to each dish using a graduated syringe
Temperature differences between locations	Confounding variable if temperature is not the independent variable	Keep all dishes at the same temperature (e.g. same incubator or room); monitor with thermometer
Fungal contamination	Seeds or seedlings may die; reduces sample size	Use clean equipment; sterilise cotton wool if possible; replace contaminated dishes
Uneven light exposure within a dish	Seedlings on the edge may receive different light from those in the centre	Rotate dishes daily; use diffused light source

Exam Tip: Always clearly identify the **independent variable** (what you are changing), **dependent variable** (what you are measuring), and **control variables** (what you keep constant). Use a large sample size (many seeds per dish, many replicates) to improve reliability. State a specific **statistical test** you could use to analyse the results and explain what it tests (e.g. "I would use a Student's t-test to determine if there is a significant difference in mean shoot height between the two light conditions").

RP12: Immobilised Enzyme Investigation

Aim: Investigate the use of immobilised enzymes (e.g. lactase immobilised in alginate beads) and compare their effectiveness to free enzymes.

Biology link: Enzyme technology and biotechnology (Topic 8). Immobilised enzymes are widely used in industry because they can be reused, are more stable, and do not contaminate the product. This practical demonstrates enzyme immobilisation by entrapment in alginate beads and its application in producing lactose-free milk.

Apparatus

- Sodium alginate solution (2-3%), calcium chloride solution (1-2%)
- Enzyme: lactase solution (or other enzyme e.g. catalase, amylase)
- Syringe (10 cm^3, without needle) or pipette for forming beads
- Beakers, sieve or tea strainer, glass wool
- Syringe barrel (10-20 cm^3) as a column, clamp stand
- Substrate: full-fat or semi-skimmed milk (contains lactose)
- Glucose test strips (clinistix) or glucose oxidase strips for quantitative measurement; or Benedict's reagent + colorimeter

Method

1. **Making the beads:** Mix equal volumes of enzyme solution and sodium alginate solution thoroughly (the enzyme is trapped within the alginate matrix).
2. Draw the mixture into a syringe. Slowly drip the mixture from the syringe into a beaker of **calcium chloride solution**. The alginate reacts with Ca^{2+} ions to form insoluble calcium alginate — creating firm, spherical beads with enzyme trapped inside.
3. Leave the beads in $CaCl_2$ for **5-10 minutes to harden** fully. Rinse with distilled water.
4. **Setting up the column:** Place a small piece of glass wool at the bottom of a syringe barrel (to retain the beads). Pack the beads into the column. Clamp the column vertically.
5. **Running the experiment:** Pour milk (substrate) slowly through the column. Collect the liquid emerging from the bottom (the product).

6. Test the collected product for **glucose** using glucose test strips or quantitative Benedict's test (glucose is a product of lactose hydrolysis: lactose → glucose + galactose).
7. **Compare** the product from immobilised enzyme with a control using free (dissolved) lactase mixed directly with milk for the same duration.
8. Investigate **reusability**: pass fresh milk through the column multiple times and test the product each time. Compare glucose concentration across successive runs.

Safety

- Calcium chloride is an irritant — wear eye protection and avoid skin contact.
- Wash hands after handling enzymes and chemicals.
- Do not consume any products — they are not food-safe laboratory preparations.

Analysis

- Compare **glucose concentration** in the product from immobilised vs free enzyme — both should convert lactose to glucose, but the immobilised enzyme may give a lower initial conversion rate (substrate must diffuse through the alginate to reach the enzyme).
- Test **reusability**: pass substrate through the column 5+ times. The immobilised enzyme should still be active after multiple uses, while a new batch of free enzyme would be needed each time.
- Investigate the effect of **temperature** on immobilised vs free enzyme: immobilised enzymes are often more thermally stable because the alginate matrix restricts molecular movement, reducing the chance of denaturation.

Advantages of Immobilised Enzymes (Key Exam Content)

Advantage	Explanation
Reusable	Enzyme stays in the column and can process many batches of substrate — dramatically reduces cost in industrial processes
Product not contaminated	Enzyme stays in the beads/column; product collected separately — no need for costly enzyme removal/purification steps
Greater stability	Alginate matrix may protect enzyme from temperature and pH changes; enzyme is more resistant to denaturation

Continuous processing	Substrate can be passed continuously through a column — ideal for industrial-scale production (no need for batch processing)

Sources of Error and Improvements

Error	Effect	Improvement
Beads of uneven size	Variable surface area; inconsistent substrate contact; uneven reaction rates	Keep syringe at a constant height above the $CaCl_2$; drip at a steady rate; practise technique
Substrate flows through column too quickly	Insufficient contact time between substrate and enzyme; low product yield	Reduce flow rate (partially clamp outlet tubing); use a longer column with more beads
Air gaps in column between beads	Substrate bypasses beads through air channels	Pack column carefully; tap gently to settle beads; add beads under water to avoid trapping air
Enzyme leaching out of beads over time	Decreasing activity with each successive run	Ensure beads are well-hardened (leave in $CaCl_2$ long enough); increase alginate concentration for firmer beads
Comparing free and immobilised enzyme without standardising enzyme concentration	Unfair comparison	Use the same total amount of enzyme in both; same substrate volume and concentration

Exam Tip: Be able to state the advantages of immobilised enzymes over free enzymes: (1) they can be **reused**, reducing costs; (2) the product is **not contaminated** with enzyme; (3) they are often **more thermally stable**; (4) they allow **continuous-flow processing**. The most common industrial example is **immobilised lactase** used to produce lactose-free dairy products for people with lactose intolerance. Other examples: immobilised glucose isomerase

(converts glucose to fructose for "high-fructose corn syrup"), immobilised penicillin acylase (modifies penicillin to produce semi-synthetic antibiotics).